

Point of Care: Modernizing Bedside Charting for Rural Hospitals with HL7v2 Interoperability



Software Requirements Specification

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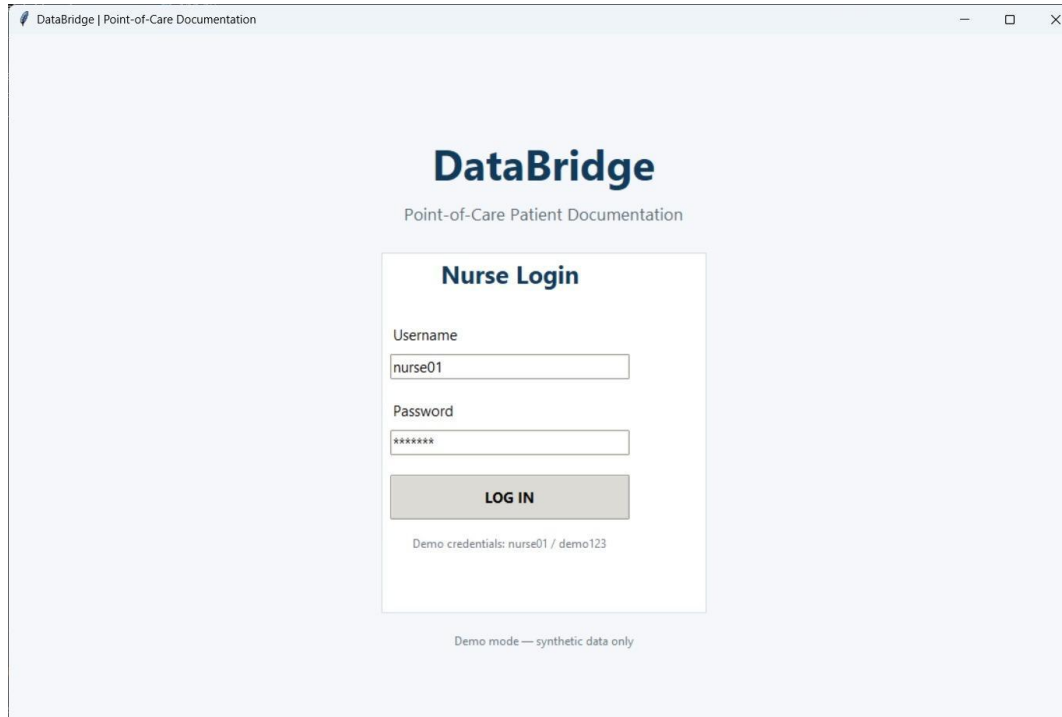
Introduction

- Purpose: Defines the requirements for DataBridge, a browser-based bedside documentation system for rural hospitals on legacy MEDITECH platforms.
- Audience: Development team, Project Mentor and Course Evaluators
- Scope: Nurses enter vitals and notes on a tablet instead of paper. Each entry is converted into an HL7v2 message and sent to a simulated EHR receiver. Goal is to reduce delays and errors from double documentation.

Overall Description

- Product Perspective: A lightweight layer between the nurse and the hospital's existing EHR. Made up of a table frontend, a backed that validates and converts data and a receiver.
- User Classes:
 - Nurse: Primary user, documents vitals for assigned patients
 - Charge Nurse: Views the full unit census
 - Administrator: Manages users and reviews audit logs.
- Operating Environment: Web browser on iPad or desktop. Backend in Python with FastAPI
- Constraints: No live hospital access, synthetic data only, HL7v2 used because legacy MEDITECH uses it.
- Assumptions: Relies on open-source HL7v2 libraries, Synthea for synthetic data, and free tier hosting.
- User Documentation: Short user guide for nurses and an administrator guide.

External Interface Requirements



The screenshot shows a web browser window titled "DataBridge | Point-of-Care Documentation". The main heading is "DataBridge" in a large, bold, dark blue font, with the subtitle "Point-of-Care Patient Documentation" below it. A central white box contains the "Nurse Login" section. It includes a "Username" label and a text input field containing "nurse01". Below that is a "Password" label and a text input field filled with asterisks. A grey "LOG IN" button is positioned below the password field. Underneath the button, it says "Demo credentials: nurse01 / demo123". At the very bottom of the page, a small note reads "Demo mode — synthetic data only".

DataBridge | Point-of-Care Documentation

DataBridge

Point-of-Care Patient Documentation

Nurse Login

Username
nurse01

Password

LOG IN

Demo credentials: nurse01 / demo123

Demo mode — synthetic data only

DataBridge | Point-of-Care Documentation

DataBridge

Select a patient to document

nurse01 | Nurse

Active Patient Census

Choose a synthetic patient record to begin bedside documentation.

Patient	Room	Status	Date of Birth
Kaitlin McAlister (P1001)	201	Stable	1991-07-07
Angela Peterson (P1002)	202	Alert	1975-12-30
Veronica Guy (P1003)	203	Stable	2006-08-17
Kaiden Radford (P1004)	204	Stable	2005-01-10
James McDaniel (P1005)	205	Monitoring	2005-08-27

DataBridge | Point-of-Care Documentation

DataBridge

Patient P1001 • Room 201

nurse01 | Nurse

Kaitlin McAlister

Room 201 • Status: Stable

Vital Signs

Temperature °F

Heart Rate bpm

Blood Pressure mmHg

Respiratory Rate /min

SpO₂ %

Observations

Nursing Notes

← BACK TO CENSUS

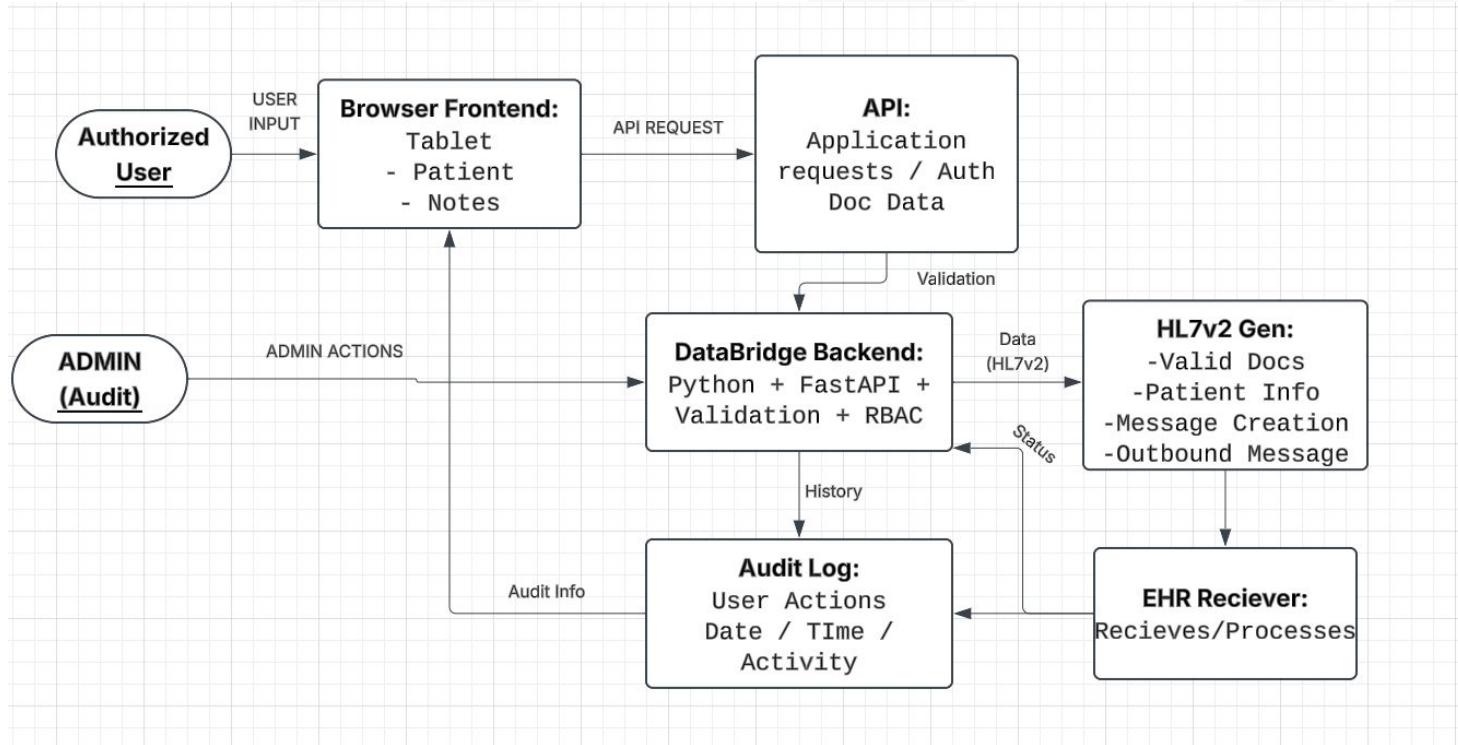
SUBMIT DOCUMENTATION

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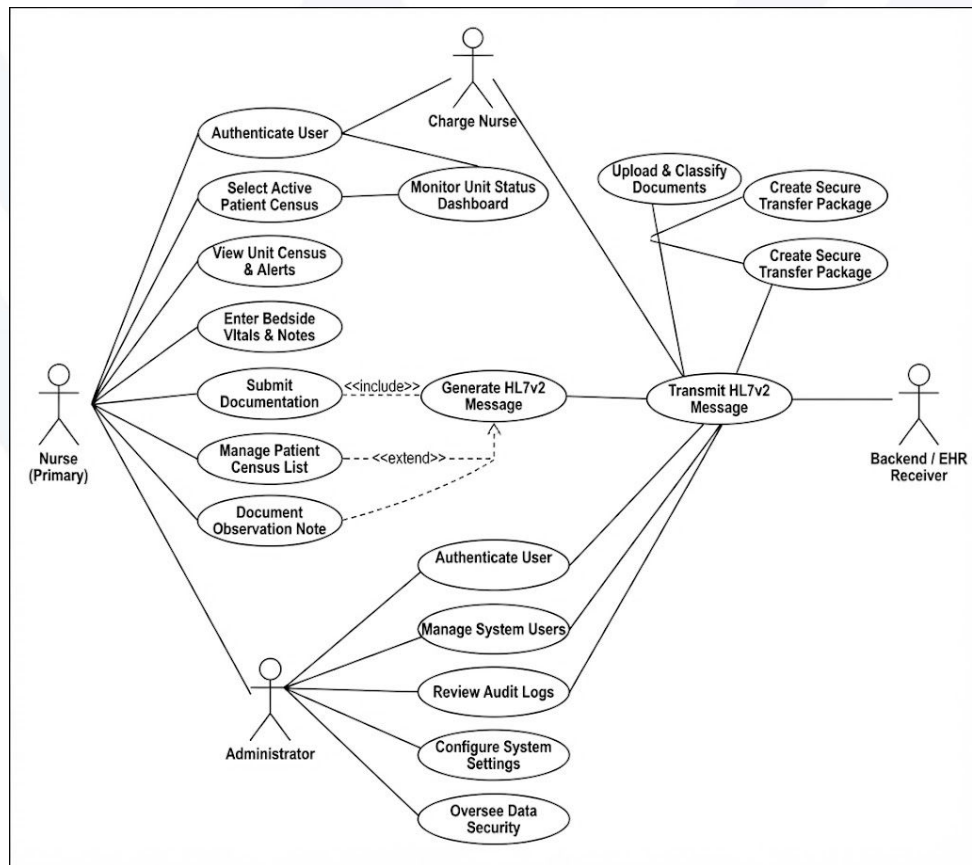
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System Features

- High Priority:
 - User Authentication
 - Role-Based Access Control
 - Bedside Documentation
 - Backend Validation and Processing
 - HL7v2 Message Generation
 - Simulated EHR Receiver
- Medium Priority:
 - Audit Logging
 - Document Upload, Classification, and OCR
 - Secure Document Transfer



Use-Case Main Structure



Other Nonfunctional Requirements

- Performance: Responsive interface: feedback after submission, full workflow runs without manual steps.
- Safety: No diagnoses or treatment recommendations. Invalid data is blocked. Synthetic Data only.
- Security: Password plus MFA, role-based access with least privileged, encryption in transit and at rest, audit logging, expiring securing sharing links. HIPAA-aligned but not HIPAA-certified.
- Quality Attributes: Easy to use on a tablet, reliable error handling, HL7v2 interoperability, separated components, testable with synthetic data.
- Business Rules: Users must log in, every user has a role, users only access what their role allows, documentation is tied to the user and patient, simulated receiver used instead of live EHR

Thank you for listening!